

- 2 (a) (i) It states that the total momentum of a system of bodies is constant provided no external resultant force acts on the system. **B1**

MC:	<i>This principle applies not only to collision problems e.g. applies during alpha decay as well. In addition, students must explain the meaning of conserve in their answers. It is also not necessary to mention that the system is isolated as no external force would imply such as.</i>
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- (ii) By conservation of momentum, **C1**
 $(2.4 \times 3.0) - (1.2 \times 2.0) = 3.6v$
 $v = 1.3 \text{ m s}^{-1}$ **A1**

- (iii) Initial total kinetic energy calculated correctly ($10.8 + 2.4 = \mathbf{13.2 \text{ J}}$) and
 Final total kinetic energy calculated correctly (**3.2 J**) **M1**
 Since total kinetic energy of system is not constant, the collision is inelastic. **B1**

OR

relative speed of approach = 5.0 m s^{-1} , relative speed of separation = 0 m s^{-1} **M1**

Since relative speed of approach is not equal to relative speed of separation, the collision is inelastic. **B1**

MC:	<i>As it is a 'show' type of question, answers need to be written in full with no short form used, i.e. KE is not accepted but have to spell out as kinetic energy.</i>
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- (b) (i) By Newton's second law, there is a net downward force by the rotor on the air to increase the momentum of air. **B1**

By Newton's third law, the air exerts a force of equal magnitude but upward direction (i.e. lift force) on the rotor. **B1**

Since the helicopter is hovering, by Newton's first law, the net force on helicopter is zero, that is, lift force = weight of helicopter. **B1**

- (ii) volume of air displaced downwards V
 = cross-section area of air $\times$ distance moved by the air in 5 seconds
 $= \pi(5.0^2)(12)(5.0)$
 mass of air displaced downwards in 5 seconds
 = volume of air $\times$ density of air
 $= \pi(5.0^2)(12)(5.0)(1.3)$ **C1**
 $= 6100 \text{ kg (2 s.f.)}$ **A0**

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MC:	<i>As it is a 'show' type of question, answers need to be written in full, before any substitution of values. Any symbols used must be defined.</i>
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(iii) By Newton's second law, $F = \text{rate of change of momentum}$

$$= \frac{6000}{5.0}(12) \quad \text{C1}$$

$$= 1.4 \times 10^4 \text{ N} \quad \text{A1}$$

MC:	<i>Some students mistakenly took the mass of air propelled downwards as the mass of the helicopter.</i>
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3 (a) The density of fluid, ρ , the cross section area, A , of tube, the total mass, M , of